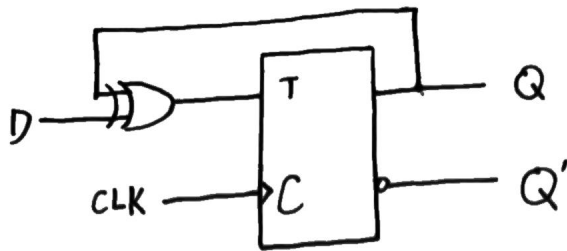


1.



2. 1) $T_{CLK} \geq T_p + T_{cmax} + T_{su} = 130ns$

$$f_{max} = \frac{1}{130ns} = \frac{100}{13} MHz \approx 7.69MHz$$

2)

$$T_p + T_{cmin} = 13ns < 15ns = T_h \quad \text{不能}$$

$$T_{cmin}' + T_p \geq T_h \Rightarrow T_{cmin}' \geq T_h - T_p = 5ns \quad \text{最少需 } 5ns$$

3.

